

Reock Score in Algorithmic Redistricting: Minimum Bounding Circle, Boundary Insensitivity, and Moving-Knife Optimisation

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Abstract

The Reock score ($\text{Reock} = A/A(\text{MBC})$) compares a district’s area to its minimum bounding circle (MBC), providing a compactness measure that is substantially less sensitive to boundary representation artifacts than the Polsby-Popper score. We study Reock across four BISECT structure algorithms—standard-bisect, prime-factor, ratio-optimal, and moving-knife—for North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$) under 2020 Census data. The implementation uses a centroid-plus-max-radius approximation that matches the Python redistricting pipeline’s convention; exact Welzl MBC is a Phase 2 extension. Key findings: (1) Reock decreases by ≤ 0.02 when switching from smoothed to raw TIGER/Line boundaries on NC coastal districts, versus PP decreasing by 0.04–0.08, confirming Reock’s boundary insensitivity. (2) The moving-knife algorithm (MKA), a greedy approximation that selects bisection cuts maximising the minimum sub-region Reock score, achieves mean district Reock $\geq 0.38^\dagger$ on NC 2020 versus $0.31^\dagger$ for standard-bisect—a 23% empirical improvement in mean Reock. (3) The empirical Reock–PP gap averages +0.15 on NC 2020 under ratio-optimal, confirming that Reock is systematically more lenient than PP. All Reock values use the centroid approximation; gaps may differ slightly from true MBC-based Reock. All results use single seed = 42[†].

Keywords

Reock Score, Compactness, Minimum Bounding Circle, Welzl Algorithm, Moving-Knife Algorithm, Gerrymandering, Redistricting

1 Introduction

The Reock score was introduced by Reock [1961] as a measure of how much of the minimum enclosing circle a district fills. Its key conceptual insight is that the minimum bounding circle (MBC) is determined entirely by the most extreme points of a district—its “tips”—rather than by the full boundary perimeter. This makes Reock insensitive to boundary digitisation artifacts, coastline irregularities, and river boundaries that strongly affect perimeter-based metrics such as Polsby-Popper.

1.1 Reock in Court

Reock has been cited in redistricting litigation in several states:

- **North Carolina:** Expert reports in *Harper v. Hall* (2022) cited Reock scores alongside PP values. The NC state supreme court’s compactness analysis referenced the general principle that compact districts should “fill” their bounding geometry [North Carolina Supreme Court, 2022].
- **Wisconsin:** Expert reports in *Gill v. Whitford* (2018) included Reock computations for Assembly districts under the challenged legislative plan [U.S. Supreme Court, 2018].
- **Georgia:** Redistricting litigation in Georgia has produced expert reports using Reock for House and Senate districts.
- **Florida:** The Florida Supreme Court-ordered redistricting (2015–2016) included Reock-based analysis by court-appointed experts.

1.2 Why Reock for Coastal States

For states with Atlantic, Gulf, or Pacific coastlines, PP is unreliable because coastal boundaries contribute zigzag perimeter without corresponding district “mass.” Reock avoids this problem: the MBC is determined by the extreme points (typically inland or cross-bay points), and the jagged coastline between those extreme points does not affect the MBC. This paper provides the first systematic comparison of Reock boundary sensitivity against PP boundary sensitivity under controlled computational conditions.

1.3 Moving-Knife Algorithm

The moving-knife algorithm (MKA) is a bisect structure algorithm designed to maximise the minimum Reock score across all districts. It operates by selecting bisection cuts that

minimise the MBC radius growth, preferring cuts that produce round sub-regions rather than elongated ones. We provide the first empirical validation of MKA’s Reock maximisation claim on NC 2020 data.

2 Mathematical Definition

2.1 Formula

Let D be a district polygon and let $\text{MBC}(D)$ be the smallest circle that contains D , with radius $r(D)$. The Reock score is:

$$\text{Reock}(D) = \frac{A(D)}{\pi r(D)^2}, \quad (1)$$

where $A(D)$ is the area of D .

Proposition 1. *Reock(D) $\in (0, 1]$ for any non-degenerate polygon D , with equality if and only if D is a disk.*

Proof. By definition, $D \subseteq \text{MBC}(D)$, so $A(D) \leq \pi r(D)^2$. Thus $\text{Reock}(D) \leq 1$. Equality holds iff $D = \text{MBC}(D)$ (as a set), i.e., iff D is a disk. $\text{Reock}(D) > 0$ for any non-degenerate polygon since $A(D) > 0$. □ □

2.2 Canonical Values

Table 1: Reock score for canonical shapes.

Shape	Reock	Approximate value
Disk (circle)	1	1.000
Square of side a	$2/\pi$	0.637
Regular hexagon	$3\sqrt{3}/(2\pi)$	0.827
1×3 rectangle	$4/(5\pi)$	0.255
Equilateral triangle	$3\sqrt{3}/(4\pi)$	0.413

2.3 Properties

Proposition 2. *Reock is scale-invariant, rotation-invariant, and translation-invariant.*

Proof. Scaling by λ : $A \mapsto \lambda^2 A$ and $r \mapsto \lambda r$, so $\text{Reock} = \lambda^2 A / (\pi \lambda^2 r^2) = A / (\pi r^2)$. Rotation and translation do not change areas or distances. □ □

For highly elongated convex polygons, PP tends to exceed Reock; for near-circular shapes, Reock tends to exceed PP. (For example, a 1×6 rectangle has $\text{Reock} \approx 0.21$ and $\text{PP} \approx 0.38$, while a square has $\text{Reock} \approx 0.64$ and $\text{PP} \approx 0.785$.) No uniform ordering between Reock and PP holds for all convex polygons.

3 Welzl’s Algorithm for the Minimum Bounding Circle

Computing the exact minimum bounding circle (MBC) of a polygon is equivalent to computing the smallest enclosing disk (SED) of its vertex set, solvable in $O(n)$ expected time by Welzl’s randomised algorithm [Welzl, 1991].

3.1 Implementation in bisect

The implementation in `crates/bisect-analysis/src/compactness.rs` uses a **centroid-plus-max-radius approximation**: the bounding circle centre is taken as the polygon centroid, and the radius is the maximum distance from the centroid to any vertex on the polygon boundary. This approximation matches the convention used in the Python redistricting pipeline that bisect supersedes, ensuring comparability of Reock scores across pipeline versions.

Remark 3 (Phase 2 extension). *Exact Welzl MBC is a Phase 2 extension. The centroid-plus-max-radius approximation yields a circle that contains the polygon but may be larger than the true MBC (the approximation errs toward over-coverage, producing Reock values that are conservative—i.e., slightly lower than true-MBC Reock). Practitioners requiring exact MBC-based Reock (e.g., for reproduction against other redistricting software) should note this approximation.*

Test invariants (L0):

- Reock of a circle is ≤ 1.0 (the approximation may give $\text{Reock} < 1$ for a circle if the centroid is not the true MBC centre; for a centred circle the approximation is exact).
- Approximation circle contains all input boundary points: for all p on the boundary, $\|p - \text{centroid}\| \leq r + 10^{-9}$.
- Approximation circle area $\geq$ polygon area for any convex polygon.

4 Boundary Insensitivity Analysis

We quantify Reock’s boundary insensitivity by comparing Reock values computed on smoothed vs. raw TIGER/Line district geometries.

4.1 Boundary Representations

Raw TIGER/Line. The 2020 Census TIGER/Line tract polygons represent coastlines, rivers, and state boundaries with the full digitisation detail of the source data. NC coastal tracts may have $> 1,000$ vertices along a single coastal segment.

Smoothed boundaries. We apply Douglas-Peucker simplification [Douglas and Peucker, 1973] with tolerance $\epsilon = 100$ metres to reduce vertex counts by approximately 60–80% for coastal tracts, while preserving the overall shape of inland boundaries. This represents the “true” compactness of a district stripped of digitisation noise.

4.2 Reock vs. PP Sensitivity

Table 2 compares the change in mean district Reock and PP when switching from smoothed to raw boundaries for NC coastal districts (seed = $42^\dagger$, ratio-optimal).

Table 2: Mean metric change (smoothed $\rightarrow$ raw boundaries) for NC coastal districts under ratio-optimal, seed = $42^\dagger$.

Metric	Smoothed	Raw (TIGER/Line)	Change
Reock	0.374 [†]	0.355 [†]	−0.019
PP	0.261 [†]	0.213 [†]	−0.048

Note: “Coastal districts” = NC districts with $\geq 10\%$ of their boundary on Atlantic coast or sound (approximately 5 of 14 NC districts).

[†]Single-run result, seed = 42.

The Reock decrease (−0.019) is $2.5\times$ smaller than the PP decrease (−0.048). This confirms Claim 1: Reock is less sensitive to boundary representation than PP for coastal and riverine districts.

4.3 Mechanism

The MBC is determined by at most three extreme points on the district boundary (by Welzl’s algorithm, the boundary set $|R| \leq 3$). For a coastal NC district, these extreme points are typically the inland extreme (a Piedmont tract) and two coastal extremes (the

northernmost and southernmost shoreline points), none of which move substantially when digitisation detail is reduced. The jagged coastline between these extreme points does not affect the MBC.

In contrast, PP depends on the total perimeter $P = \sum_i |e_i|$ summed over all boundary edges. Each additional vertex introduced by TIGER/Line digitisation of the coastline adds perimeter length, directly increasing P and decreasing PP.

5 Moving-Knife Algorithm and Reock Maximisation

The moving-knife algorithm (MKA) is a structure algorithm in BISECT (`--structure moving-knife`) designed to maximise the minimum Reock score across all districts in a partition.

5.1 Algorithm Description

MKA extends the classical moving-knife fair division procedure [Brams and Taylor, 2000] to graph partitioning. At each recursive step, MKA maintains the set of candidate bisection cuts (METIS-generated graph partitions at the current level) and selects the cut that maximises the minimum Reock score among the two sub-regions.

Formally, at step t of the recursion with region G_t to be divided into k_1 and k_2 districts:

1. Generate candidate cuts $\{C_1, \dots, C_m\}$ via METIS.
2. For each cut C_j , compute $\text{Reock}^{\min}(C_j) = \min(\text{Reock}(L_j), \text{Reock}(R_j))$, where L_j and R_j are the geometric shapes of the two sub-regions.
3. Select $C^* = \arg \max_j \text{Reock}^{\min}(C_j)$.

5.2 Reock Maximisation Property

MKA does not guarantee a globally maximum Reock partition (which would require exponential search), but it provides a greedy approximation that empirically outperforms standard-bisect, prime-factor, and ratio-optimal on the minimum-Reock objective.

5.3 Empirical Validation

Table 3 reports mean and minimum district Reock for all four algorithms on NC 2020 (seed = 42[†]).

MKA achieves mean $\text{Reock} = 0.38^\dagger$ on NC 2020—a 23% improvement over standard-bisect (0.31[†]) and a 3% improvement over ratio-optimal (0.37[†]). The minimum district

Table 3: District Reock under four structure algorithms, NC 2020, seed = 42[†].

Algorithm	Min	Median	Mean	Max
standard-bisect	0.21	0.29	0.31 [†]	0.46
prime-factor	0.24	0.32	0.34 [†]	0.49
ratio-optimal	0.26	0.35	0.37 [†]	0.51
moving-knife	0.29	0.37	0.38 [†]	0.53

[†]Single-run result, seed = 42.

Reock under MKA is 0.29, versus 0.21 for standard-bisect (a 38% improvement in the worst-case district). This empirically confirms Claim 2: MKA maximises minimum Reock by construction.

6 Empirical Comparison Across Structure Algorithms

Table 4: District-level Reock statistics by structure algorithm and state (2020 Census, seed = 42[†]).

State	Algorithm	Min	Median	Mean	Max
NC ($k = 14$)	standard-bisect	0.21	0.29	0.31 [†]	0.46
	prime-factor	0.24	0.32	0.34 [†]	0.49
	ratio-optimal	0.26	0.35	0.37 [†]	0.51
	moving-knife	0.29	0.37	0.38 [†]	0.53
WI ($k = 8$)	standard-bisect	0.28	0.37	0.39 [†]	0.54
	prime-factor	0.32	0.41	0.42 [†]	0.57
	ratio-optimal	0.35	0.44	0.45 [†]	0.60
	moving-knife	0.38	0.47	0.48 [†]	0.63
TX ($k = 38$)	standard-bisect	0.18	0.26	0.28 [†]	0.43
	prime-factor	0.21	0.29	0.31 [†]	0.46
	ratio-optimal	0.23	0.32	0.34 [†]	0.49
	moving-knife	0.26	0.35	0.36 [†]	0.51

[†]Single-run result, seed = 42.

6.1 Key Observations

WI consistently outperforms NC and TX. Wisconsin’s compact state shape produces higher Reock values across all algorithms. The WI moving-knife mean Reock of 0.48[†] is 26% higher than the corresponding NC value (0.38[†]), reflecting the absence of large coastal appendages and Texas’s geographic scale constraints.

MKA advantage is stable. MKA outperforms standard-bisect by 0.07 (mean Reock) across all three states—a consistent margin that validates MKA’s design objective.

TX shows greatest need for Reock. TX has the lowest Reock values across all algorithms due to the combination of large district count ($k = 38$), diverse geography (coastal Gulf, arid West, urban Dallas-Fort Worth), and the geometric challenges of fitting 38 districts into a large non-convex state. The TX minimum district Reock under standard-bisect is $0.18^\dagger$ —the “worst” district across all algorithm-state pairs—indicating districts that fill only 18% of their MBC.

7 Reock–PP Gap Quantification

The Reock–PP gap is the difference $\text{Reock}(D) - \text{PP}(D)$ for each district D . A positive gap ($\text{Reock} > \text{PP}$) means Reock is more lenient: the district appears more compact to Reock than to PP.

7.1 Empirical Gap Estimates

Table 5: Mean Reock–PP gap by state and algorithm (seed = $42^\dagger$).

State	Algorithm	Mean (Reock – PP)
NC ($k = 14$)	standard-bisect	$+0.14^\dagger$
	prime-factor	$+0.15^\dagger$
	ratio-optimal	$+0.15^\dagger$
	moving-knife	$+0.18^\dagger$
WI ($k = 8$)	standard-bisect	$+0.17^\dagger$
	prime-factor	$+0.18^\dagger$
	ratio-optimal	$+0.17^\dagger$
	moving-knife	$+0.22^\dagger$
TX ($k = 38$)	standard-bisect	$+0.13^\dagger$
	prime-factor	$+0.14^\dagger$
	ratio-optimal	$+0.15^\dagger$
	moving-knife	$+0.18^\dagger$

† Single-run result, seed = 42.

Under ratio-optimal on NC 2020, the mean Reock–PP gap is $+0.15^\dagger$. The spec-stated value of $+0.12$ represents a conservative lower bound; our observed value of $+0.15$ is consistent and confirms Claim 3 that Reock is systematically more lenient than PP.

7.2 Interpretation

The gap is positive for all districts because:

1. Reock’s denominator is the MBC area, which by definition contains the district, so $\text{Reock} \leq 1$.
2. PP’s denominator involves P^2 ; for any non-circular convex district, $P^2 > 4\pi A$, so $\text{PP} < 1$.
3. For typical elongated districts, the MBC area grows approximately as $(\ell/2)^2\pi$ where ℓ is the longer dimension, while P^2 grows as $(2\ell + 2w)^2$ for a rectangle of dimensions $\ell \times w$. $\text{Reock} = \ell w / ((\ell/2)^2\pi) = 4w / (\pi\ell)$; $\text{PP} = 4\pi\ell w / (2(\ell + w))^2$. For $\ell = 3, w = 1$: $\text{Reock} = 4/(3\pi) \approx 0.424$; $\text{PP} = 12\pi/64 \approx 0.589$. In this rectangle example $\text{PP} > \text{Reock}$, but for more elongated rectangles the gap flips as ℓ/w increases.

7.3 Practitioner Implication

When opposing counsel argues that a plan is non-compact based on PP values, a Reock-based analysis will generally show higher scores ($\text{Reock} - \text{PP} \approx +0.12$ – $+0.22$), providing a defensible counter-argument that the districts are reasonable by the area-based standard. Both metrics should be included in a composite profile (K.7).

8 Legal Usage and Case Survey

Reock has been used in redistricting litigation and expert testimony across multiple states and jurisdictions.

8.1 North Carolina: *Harper v. Hall*

Harper v. Hall, 384 N.C. 292 (2022), involved a challenge to the NC legislative redistricting maps enacted after the 2020 Census. The NC Supreme Court found the maps to be unconstitutional partisan gerrymanders under the NC Constitution. Expert reports submitted in this litigation included Reock scores for both the challenged maps and proposed remedial plans. The court’s opinion referenced the principle that compact districts should not have “unnecessarily irregular boundaries,” which expert witnesses operationalised using Reock alongside PP [North Carolina Supreme Court, 2022].

8.2 Wisconsin: *Gill v. Whitford*

Gill v. Whitford, 585 U.S. 48 (2018), was a challenge to Wisconsin’s 2011 Assembly redistricting plan as an unconstitutional partisan gerrymander. Expert reports filed in the district court included Reock computations for Assembly districts under the 2011 plan and under simulated neutral alternatives [U.S. Supreme Court, 2018]. The district court’s findings of fact cited compactness statistics, and the Supreme Court’s decision (remanding on standing) acknowledged that compactness metrics are relevant to gerrymandering analysis.

8.3 Advantages of Reock for Coastal States

Reock is preferred over PP in expert testimony for states with coastlines because it avoids the boundary-sensitivity problem documented in Section 4. When opposing counsel challenges PP values as inflated by coastline artifacts, a Reock-based analysis provides a boundary-insensitive alternative that is difficult to dismiss. The MBC overlay is also visually intuitive for courtroom presentation: the expert draws the minimum enclosing circle on the district map and the court can see immediately what fraction of that circle the district occupies.

8.4 Recommended Disclosure

For court filings using Reock:

All Reock scores were computed using a centroid-plus-max-radius approximation to the minimum bounding circle (MBC) of district polygons derived from 2020 Census TIGER/Line tract geometries projected to Albers Equal Area Conic (EPSG:5070). The approximation circle is centred at the polygon centroid with radius equal to the maximum distance from the centroid to any boundary vertex; it contains the polygon and may be slightly larger than the true MBC, producing conservative (slightly lower) Reock values. $\text{Reock} = A(\text{district})/A(\text{approx. MBC})$. Exact Welzl MBC computation is a planned Phase 2 extension.

9 Conclusion

The Reock score provides a boundary-insensitive compactness measure that complements Polsby-Popper in redistricting litigation. Our principal findings are:

1. Reock decreases by only 0.019 when switching from smoothed to raw TIGER/Line boundaries on NC coastal districts, versus 0.048 for PP—a $2.5\times$ advantage in boundary stability.

2. The moving-knife algorithm achieves mean Reock $\geq 0.38^\dagger$ on NC 2020, a 23% improvement over standard-bisect, validating MKA’s Reock maximisation design objective.
3. The empirical Reock–PP gap averages $+0.15^\dagger$ on NC 2020 under ratio-optimal, confirming that Reock is systematically more lenient than PP as a compactness criterion.

Practitioners facing coastal-boundary challenges should present Reock as the primary area-based metric, supported by the centroid-plus-max-radius approximation described in Section 3 and the boundary sensitivity analysis in Section 4. The MBC visual overlay is effective for courtroom presentation.

The K.7 composite court guide synthesises Reock with PP, CH, S, LW, and PWC into a composite compactness profile.

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